

Contact

Bengaluru
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Top Skills

Distributed Training
Quantization Techniques
OpenCL

Languages

Hindi (Native or Bilingual)
English (Professional Working)

Certifications

Neural Networks and Deep Learning
Supervised Machine Learning:
Regression and Classification
Structuring Machine Learning
Projects
Generative AI with Large Language
Models
Foundations: Data, Data,
Everywhere

SAMUNDER SINGH

Design Engineer at CDAC Bangalore | building llm for edge device |
vision AI | open source
Bengaluru, Karnataka, India

Summary

I'm an AI/ML design Engineer working at the intersection of LLMs, distributed systems, GPU computing, and AI infrastructure. Currently at C-DAC Bengaluru, I work on building 1B–7B SLMs, focusing on distributed training, inference optimization, quantization, and deployment. I also work on the software stack for custom GPU hardware, including OpenCL kernels, RISC-V, POCL, compiler/runtime development, and enabling PyTorch-like AI workloads on custom accelerators. Previously at Lincode, I built and deployed production ML/CV systems using PyTorch, YOLO, DETR, CLIP, TensorRT, torch.compile, and distributed training. Interested in LLM systems, GPU programming, compilers, AI accelerators, and hardware–software co-design.

Experience

CDACINDIA

Design Engineer
July 2026 - Present (2 months)
Bengaluru

- Building end-to-end infrastructure for distributed SLM training and inference, covering training pipelines, inference optimization, and model quantization for in-house 1B–7B parameter models.

- Developing the software stack for a custom GPU accelerator, working at the compiler/runtime level with OpenCL kernels, RISC-V POC, and POCL to enable efficient AI/LLM workloads on the in-house GPU architecture.

- Enabling PyTorch-like AI frameworks and LLM inference on custom hardware, bridging the gap from low-level compiler and kernel development to high-level AI workloads and optimizing the complete hardware–software stack.

Lincode

2 years 3 months

Associate Data Scientist

September 2024 - July 2026 (1 year 11 months)

Bengaluru

- Built a synthetic data pipeline with a structured defect library — covering defect generation, augmentation, and dataset versioning — to handle rare-class imbalance in industrial inspection models.
 - Designed a self-supervised auto-labeling algorithm and a CLIP-based unsupervised pipeline for anomaly detection and data visualization.
 - Trained and deployed computer vision models (YOLOv8, DETR, CLIP) for defect detection; optimized inference using TensorRT and torch.compile for edge and cloud serving.
 - Scaled distributed model training across multi-GPU and multi-host clusters using PyTorch DDP.
 - Led end-to-end integration of segmentation and OCR modules using PaddleOCR into real-time inspection workflows.
 - Integrated multi-bucket cloud storage (AWS S3, DigitalOcean Spaces, Storj) into the product for dataset and model artifact management.
 - Built an MCP server and custom AI tools to enable LLM-driven automation of internal workflows.
- Enhanced Django-MongoDB backend for scalable ML product deployment; improved system reliability and modularity.
- Applied K-Means and DBSCAN clustering for unsupervised defect pattern discovery on unlabeled production data.
 - Used Grad-CAM and SHAP to surface model explanations for non-technical stakeholders.

Data Science Intern

May 2024 - September 2024 (5 months)

United States

- Developed and integrated object detection models (YOLO, DETR) into the product pipeline, improving real-time inspection accuracy.

- Built robust image augmentation pipelines to enhance model generalization across diverse manufacturing scenarios.
- Contributed to Django-based backend development, fixing bugs, optimizing API performance, and integrating ML models.
- Assisted in building a segmentation and OCR pipeline using PaddleOCR, successfully deploying it in production environments.

Unify

1 year

Top contributor

February 2024 - May 2024 (4 months)

- Collaborating on test case fixes and critical tasks with the Ivy team, involving deep dives into transpiler and compiler code, and reviewing pull requests.
- developed a chatbot to compare and rank the top LLM with a global leaderboard user can ask query to two llm same time can vote the winner
- contributed in handling the complex dtype at tensorflow , jax and paddlepaddle backend improved the some SHLO in xla for complex dtype.

Open Source Developer

June 2023 - February 2024 (9 months)

London, England, United Kingdom

- open source a machine learning tool that helps optimize models, converts code between frameworks, and allows writing code independent of specific ML frameworks.
- Contributed with across JAX, NumPy, PaddlePaddle, TensorFlow, PyTorch, and Ivy function API, enhancing machine learning frontend APIs
- Implemented a range of non-linear activations in Ivy's API (e.g., HardTanh, Softshrink) for versatile neural network customization. Contributed to Paddle's frontend with functions like Reciprocal, Tanh, and Remainder. Enhanced JAX by integrating Std, Var, ELU Activation for statistical analysis and varied activations.

<https://aaranyak.org/>

Computer Vision Research

February 2023 - June 2023 (5 months)

Guwahati, Assam, India

- Developed precise Rhino height estimation models and a robust 98% accurate detection pipeline.
- Led model development, optimized depth estimation for accurate height predictions, and designed the Rhino identification pipeline.
- Achieved <2% error in Rhino height estimation, 98% accurate Rhino detection, and utilized depth estimation for improved accuracy.

SmartKnower

Data Scientist intern

April 2021 - June 2021 (3 months)

India

- Created and deployed an interactive Tableau dashboard presenting COVID-19 death and survival
- Designed intuitive interface, improving data exploration by 40%. Implemented robust email filtering achieving 98% precision, 95% recall, F1 score of 0.96 for enhanced user experience.
- Interactive COVID-19 Dashboard: Enabled exploration of regional COVID-19 trends via an intuitive Tableau interface. Accurate Spam Email Classifier: Leveraged XGBoost and SVM algorithms for 98% precision, reducing spam by 60%. Enhanced Email Filtering: Achieved 95% recall and an F1 score of 0.96, significantly improving email filtering accuracy and user satisfaction.

Education

Tezpur University

Bachelor of Technology - BTech, Electrical, Electronics and Communications Engineering · (2020 - 2024)

S.B.M senior secondary school ,suratgarh , rajasthan

High School Diploma · (2016 - 2018)